ZX-1 - AI for Smart Sensing

AI 7793 - Section W01 - Spring 2026

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Website: `https://mcmcnair2019-sys.github.io/AI-Heartbeat-Anomaly-Detection`
GitHub: `https://github.com/mcmcnair2019-sys/AI-Heartbeat-Anomaly-Detection`
Video: `https://youtu.be/4-KwnCRYee4`

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Introduction

This project, AI For Smart Sensing (ZX-1), builds an end-to-end pipeline for classifying ECG heartbeat signals using Large Language Models (LLMs) enhanced with Retrieval-Augmented Generation (RAG). The system is designed to demonstrate how IoT sensor data, which LLMs typically struggle to reason about, can be made interpretable through preprocessing, knowledge retrieval, and structured prompting.Traditional LLMs excel at natural language but perform poorly on raw numerical time-series data. This project bridges that gap by converting ECG signals into text representations, extracting statistical features, and retrieving relevant medical knowledge and reasoning demonstrations before asking the LLM to classify each heartbeat.

Goal

Classify ECG heartbeats from the MIT-BIH Arrhythmia Dataset as either Normal beats or Premature Ventricular Contractions (PVC) using a RAG-enhanced LLM pipeline, and demonstrate measurable improvement over a naive baseline that sends raw data directly to the LLM.

Requirements

- Load and preprocess ECG sensor data from the MIT-BIH Arrhythmia Dataset
- Extract heartbeat windows and resample to a consistent length
- Convert numerical ECG signals into text format consumable by an LLM
- Build a knowledgebase of domain documents and few-shot reasoning demonstrations
- Implement hybrid retrieval (keyword + semantic embedding) over the knowledgebase
- Apply reranking to improve retrieval precision
- Construct structured prompts containing signal data, features, and retrieved context
- Use an LLM to generate step-by-step reasoning and a classification label
- Evaluate accuracy on a balanced test set of 100 samples
- Compare performance against a naive baseline with no retrieval

Analysis

Raw IoT sensor data presents three core challenges for LLM-based reasoning. First, LLMs are trained primarily on text and have no inherent understanding of numerical sequences. Second, without domain-specific knowledge, the model cannot connect signal features to clinical meaning. Third, time-series data lacks the linguistic structure that LLMs rely on for reasoning.

Dataset

The MIT-BIH Arrhythmia Database contains 48 half-hour ECG recordings sampled at 360Hz. For this project, 500 Normal beats and 500 PVC beats were extracted, providing a balanced binary classification dataset of 1,000 samples. Each beat was extracted as a 0.8 second window centered on the annotated beat position.

Three features were computed per beat to supplement the raw signal text provided to the LLM. Mean was computed to determine the average signal amplitude, it is meant to determine when the baseline readings shift. Variance was also computed because PVC beats tend to show higher variance. FFT mean magnitude was then computed to capture the complexity of the waveform shape.

Development

Data Preprocessing

ECG records were loaded using the WFDB library. Each record was processed to extract the MLII lead signal. Beat annotations provided by the MIT-BIH dataset were used to center 0.8-second windows around each annotated beat. Windows were resampled from the native 360Hz to 72Hz using scipy.signal.resample(), producing exactly 57 samples per beat regardless of source record sampling rate.

Knowledgebase Design

The knowledgebase contains two types of documents. Domain documents provide medical background on ECG morphology, PVC characteristics, and signal interpretation. Demonstration examples provide complete few-shot reasoning traces, each containing a signal description, extracted features, step-by-step reasoning, and a ground truth label

RAG

The retrieval pipeline operates in two stages. First, hybrid retrieval combines normalized BM25 keyword scores and cosine embedding similarity scores at equal weight (50/50) to produce a ranked list of top-5 candidate chunks. Second, a cross-encoder reranker scores each (query, chunk) pair jointly and selects the top-2 most relevant chunks for injection into the prompt.

Prompt Engineering

Each prompt includes four components: a role definition of cardiologist, the serialized ECG signal, the three extracted features, and the retrieved knowledge context. The LLM is instructed to produce step-by-step reasoning followed by a structured output containing LABEL and CONFIDENCE fields that can be parsed programmatically.

Comparison

Three system configurations were evaluated on the same 100-sample test set (50 Normal, 50 PVC) to measure the contribution of each pipeline component:

- Baseline: No retrieval just raw signal and features only
- RAG only: Embedding-only retrieval, no reranker
- Full pipeline: Hybrid retrieval + CrossEncoder reranker

Dashboard

An interactive web dashboard was built using HTML and JavaScript, hosted publicly on GitHub Pages. The site has two main sections. The first is a pre-loaded demo that allows visitors to select from four real ECG beats extracted from the MIT-BIH dataset, two Normal and two PVC. Selecting a beat displays the ECG waveform, extracted features, and simulates the full RAG pipeline step by step, showing the hybrid retrieval process, the reranked knowledge chunks, and the LLM reasoning output with a final classification and confidence score. The second section allows users to upload their own ECG signal file in CSV or TXT format. The uploaded signal is plotted as a waveform, features are computed, and the full pipeline runs via a local API, returning the retrieved chunks, LLM reasoning, and prediction. The website also provides navigation links to the final report PDF, GitHub repository, and presentation video."

Design

Stage	Component	Purpose
Data	MIT-BIH Dataset + WFDB	Load raw ECG records and beat annotations
Data	scipy.signal.resample	Normalize all windows to 57 samples
Data	Feature Extraction	Compute mean, variance, FFT magnitude

Data	Signal Serialization	Convert waveform to comma-separated text
Knowledge	Domain Docs (10)	ECG medical background knowledge
Knowledge	Demo Examples (20)	Few-shot reasoning demonstrations
Knowledge	GTE-Large Embeddings	Dense vector representations of chunks
Knowledge	BM25 Index	Keyword-based sparse retrieval index
Retrieval	Hybrid Retrieve	Combine BM25 + cosine similarity scores
Retrieval	BGE Reranker	Cross-encoder relevance scoring
Reasoning	Prompt Construction	Inject signal + features + context
Reasoning	LLM (LLama)	Generate reasoning + classification
Evaluation	parse_label()	Extract prediction from LLM response
Evaluation	Accuracy Metrics	Overall + per-class accuracy comparison

Technology Stack

Library	Purpose
wfdb	Read MIT-BIH ECG records
scipy	Signal resampling
sentence-transformers	GTE-Large embedding model
Sentence-transformers	Cross Encoder
transformers	HuggingFace LLM pipeline

HuggingFace	LLM
html	Interactive dashboard
rank_bm25	BM25 keyword retrieval
Pandas / numpy	Data processing
matplotlib	ECG visualization

GitHub

All source code is maintained in a public GitHub repository at:
https://github.com/mcmcnair2019-sys/AI-Heartbeat-Anomaly-Detection
The repository contains the main pipeline script, knowledgebase documents, demo examples, and the project website
Key files in the repository:

- `heartbeat_anomaly.py` - main pipeline (data loading, retrieval, LLM classification, evaluation)
- `index.html` - public project website with interactive demo
- `Docs/` - domain knowledge text files used in the knowledgebase
- `Demos/` - few-shot reasoning demonstration files
- `README.md` - project overview and setup instructions

The dataset (MIT-BIH Arrhythmia Database) is not included in the repository due to file size limits. It can be downloaded from https://physionet.org/content/mitdb/1.0.0/ and placed in the project Google Drive folder before running

Summary

This project successfully demonstrated that Retrieval-Augmented Generation can significantly improve LLM performance on IoT sensor data classification tasks. By converting raw ECG signals into text, building a structured knowledgebase, and using hybrid retrieval with reranking, the system achieved measurably better classification accuracy than sending raw data directly to an LLM.

Evaluation

The baseline result of 49% overall accuracy with 0% PVC detection reveals the core problem this project addresses: without domain knowledge, the LLM defaults to predicting Normal for every sample. Since the test set is balanced 50/50, this produces near-random overall accuracy and complete failure on the abnormal class. This confirms that raw sensor data alone is insufficient for LLM-based classification.

Adding RAG retrieval produces the largest single improvement in the pipeline, raising overall accuracy from 49% to 80% and PVC detection from 0% to 66%. This demonstrates that providing the LLM with relevant medical knowledge and reasoning demonstrations is the most critical factor in enabling correct classification of abnormal beats. The full pipeline with hybrid retrieval and cross-encoder reranking improves further to 84% overall and 72% PVC accuracy. The additional 4 percentage points over RAG-only confirm that better context selection by retrieving more relevant chunks through BM25 + embedding combination and then reranking, provides benefits beyond basic semantic retrieval alone.

One observation is that Normal accuracy decreases slightly from the baseline (98%) to the full pipeline (96%). This is an expected tradeoff: adding PVC-focused retrieval context occasionally causes the model to second-guess borderline normal beats. This is a known limitation of few-shot RAG approaches and is an area for future improvement.

Achievements

- End-to-end pipeline from raw ECG sensor data to LLM classification
- 35% improvement in overall accuracy over the naive baseline (49% to 84%)
- 72% PVC detection compared to 0% with no retrieval
- Hybrid BM25 + embedding retrieval outperforms embedding-only retrieval
- Cross-encoder reranking adds further improvement over basic RAG
- Interactive website demo
- System architecture generalizable to other IoT domains

Challenges

- Windows memory limitations prevented loading the BGE reranker locally, resolved by migrating to Google Colab and swapping to CrossEncoder
- FlagEmbedding version conflicts with newer transformers required a library swap
- LLM response parsing requires robust regex fallback since models don't always follow formatting instructions

- Evaluation is slow on CPU
- Normal accuracy trades off slightly when PVC-focused context is added to the retrieval pool

Future Work

- Expand classification to additional arrhythmia classes beyond Normal and PVC
- Use a larger LLM with stronger reasoning capability to improve PVC accuracy further
- Investigate retrieval strategies that preserve Normal accuracy while improving PVC detection
- Add real-time streaming ECG input for live beat-by-beat classification
- Extend the framework to other IoT domains such as temperature anomaly detection, motion analysis, or energy usage classification